

INNOVATIONS IN THE FIELD OF AGRICULTURE

New Crops Introduced, New Techniques of Irrigation etc.

Objectives

Introduction

Significance of Agriculture in Medieval India

Condition of Agriculture

Crop Pattern

Food crops

Cash crops

Spices

Fruits and Vegetables:

New Crops

Conclusion

Overview of Medieval Indian Agriculture

- ▶ **Significance:** Agriculture was the backbone of the economy, influencing history and the financial strength of the exchequer.
- ▶ **Technological Status:** Agricultural technology was comparable to that of Europe and other parts of the world.
- ▶ **Methodology:** While ancient methods persisted, significant progress was made through the introduction of new crops and improved irrigation.
- ▶ **Government Role:** Rulers improved irrigation, supplied seeds, and solved cattle problems to boost productivity and revenue

Major Food Crops

- ▶ **Wheat:** A primary spring crop grown in regions like Agra, Delhi, and Lahore. Kabul wheat was black; Qandhar wheat was white.
- ▶ **Rice:** Required high temperatures and abundant water.
 - ▶ Shali rice was a superior fragrant variety; **Munji** rice was highly priced.
- ▶ **Millets & Pulses:**
 - ▶ **Millets:** Cheap grains like Jowar and Bajra, grown in poor soil/dry regions.
 - ▶ **Pulses:** Major protein source (moong, moth, mash, etc.), largely grown in Bihar and Oudh.

Cash Crops and Global Trade

- ▶ **Indigo:** Used for blue dye; best qualities came from Bayana (Agra) and Sarkhej (Ahmedabad).
- ▶ **Sugarcane:** Widespread by the 14th century; Bengal produced the finest quality.
- ▶ **Cotton:** Chiefly grown in Khandesh and Berar; exported to Arabia, Burma, and Africa.
- ▶ **Opium:** Manufactured from poppy seeds in Bihar and Malwa; used as an intoxicant and medicine.
- ▶ **Silk:** Sericulture became widespread during the Mughal period, with Bengal as the main hub.

Introduction of New Crops (16th-17th Century)

- ▶ **Portuguese Contributions:** Introduced tobacco, pineapple, papaya, cashew nut, and potato.
- ▶ **Global Exchange:**
 - ▶ **Cashew:** Introduced from South America.
 - ▶ **Coffee:** Introduced shortly after Akbar's reign.
 - ▶ **Vegetables:** Chilies, tomatoes, and potatoes appeared in the late medieval period.
- ▶ **Horticultural Advances:** Jesuits in Goa introduced systematic **mango grafting** in the late 16th century

Irrigation Technology

- ▶ **Traditional Methods:** Included wells (*kupa*), tanks (*tadaga*), and step-wells (*vapi/baoli*).
- ▶ **Water-Lifting Devices:**
 - ▶ **Araghatta (Persian Wheel):** Used for lifting water from deep wells, especially in Rajasthan and Gujarat.
 - ▶ **Dhenkli & Charas:** Manual and animal-powered leather bucket systems
- ▶ **Canal Systems:**
 - ▶ **Firuz Shah Tughlaq:** Built a massive network of canals and 1,200 fruit orchards.
 - ▶ **Shahjahan:** Constructed the *Nahr Faiz* to carry water from the Yamuna.

Irrigational Technology - Infrastructure

► Canal Systems:

- Ghiyassuddin Tughlaq and Firuz Shah Tughlaq built extensive canal networks.
- The *Nahr Faiz* (Yamuna canal) was built during Shahjahan's reign to irrigate the Haryana tract.

► Regional Trends:

- **Kashmir:** King Avantivarman's minister, Suyya, dammed the Jhelum to prevent floods and improve irrigation.
- **South India:** Depended heavily on tank irrigation (e.g., Anantraj Sagar tank) with complex sluice systems for water regulation.
- **Western India:** Focused on deep wells and step-wells (*baolis*) to combat water scarcity.

अरागुहत्त (Persian Wheel)



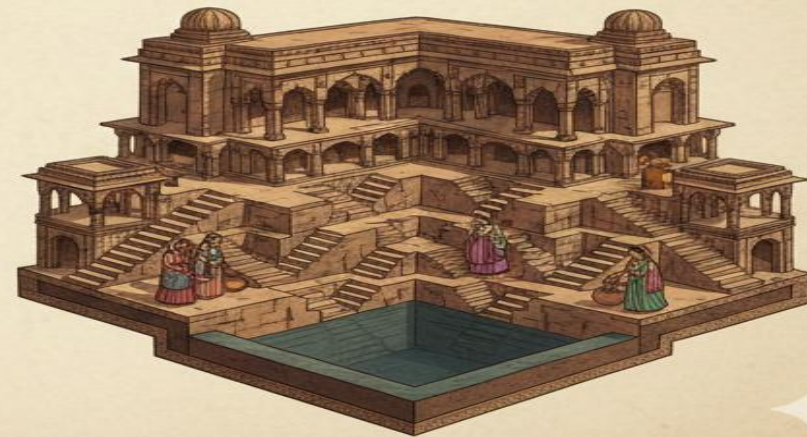
देनली
(Lever-Balance System)



धारस (Animal-Pulled Leather bag)



Step-Well (Baoli/Vapi)



Conclusion

- ▶ **Technological Fusion:** The medieval era blended traditional ancient methods with new foreign introductions (crops and tools).
- ▶ **Economic Impact:** Agricultural innovation was directly responsible for increased revenue and the survival of vast empires.
- ▶ **Solid Foundation:** By the end of this period, agriculture was stabilized through land measurement, classification, and improved hydraulic engineering.